

Table 2: Non-Functional Requirements

Attribute	Requirement
Performance	<2s response time for critical features
Scalability	Horizontal scaling with Kubernetes
Security	AES-256 encryption, PCI DSS compliance
Reliability	99.9% uptime
Usability	Simple, intuitive, accessible UI

4.2. System Architecture

The application follows a microservices architecture for flexibility and independent scaling.

Components :

1. Client Layer (React Native App)

- Cross-platform mobile app for iOS/Android.
- Handles authentication, booking, tracking, and payments.

2. API Gateway

- Single entry point for client requests.
- Routes requests to appropriate microservices.

3. Microservices

- User Service: Authentication & profile management.
- Booking Service: Booking lifecycle management.
- ML Service: Runs AI models for package recommendation & pricing.
- Chatbot Service: Rasa-based NLP system.
- Location Service: Real-time GPS tracking.